Enhancing Video-LLM Reasoning via Agent-of-Thoughts Distillation

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Abstract

Video question answering increasingly requires multi-step reasoning over long temporal spans, object interactions, and causal structure, yet end-to-end Video-LLMs remain hard to interpret and often fail on queries that demand explicit spatial-temporal grounding. We present Agent-of-Thoughts Distillation (AoTD), a framework that transfers tool-grounded reasoning from an agent system into a single generative Video-LLM. AoTDdecomposes each query into executable sub-tasks, solves them with specialized models, converts verified execution traces into natural-language rationales, filters low-quality chains of thought, and instruction-tunes a Video-LLM with both direct-answer and rationale-generation targets. Across multiple-choice and open-ended VideoQA benchmarks, AoTDconsistently improves a LLaVA-NeXT-Video 7B backbone over matched instruction tuning, raising accuracy from 53.4 to 55.6 on MVBench, from 72.2 to 74.3 on STAR, and from 47.1/3.1 to 53.9/3.4 on ANetQA, while reducing inference latency from 47.93 s to 10.58 s relative to the full agent system. The distilled model also produces grounded rationales with 21.7 temporal IoU, 34.0 temporal recall, and 45.2 spatial IoU, showing that part of the agent's spatial-temporal reasoning can be internalized at test time.

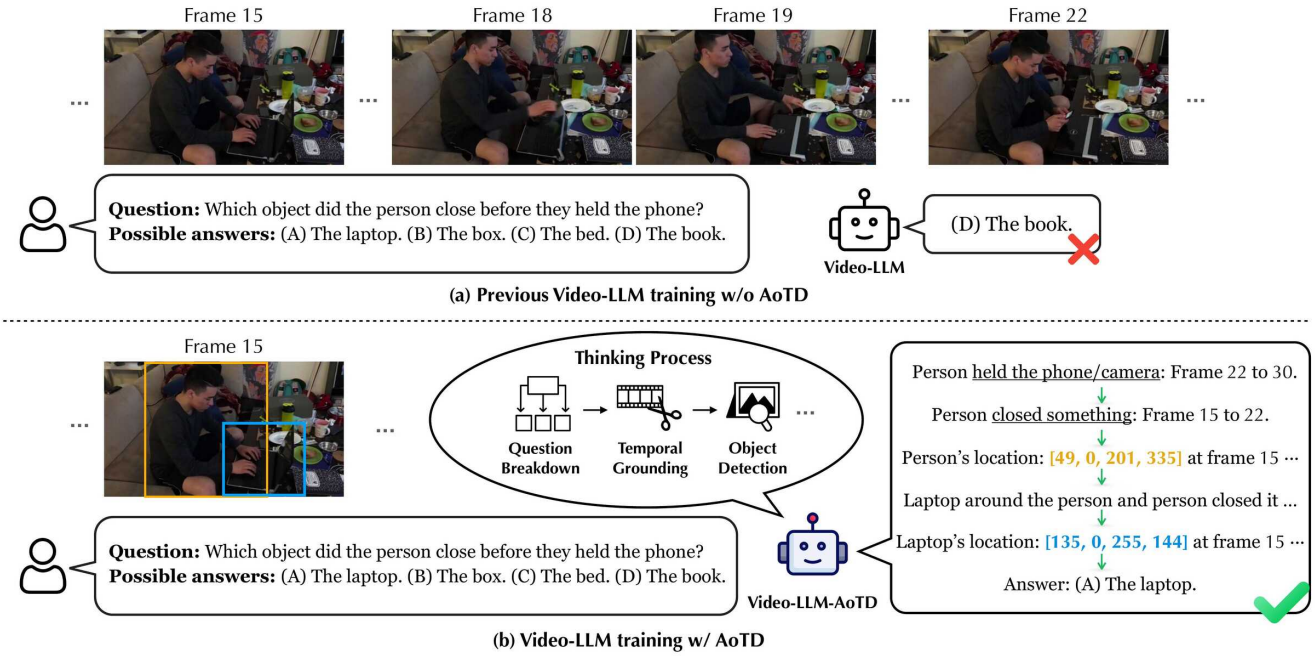

Figure 1. Our method, **AoTD**, distills multi-step reasoning and spatial-temporal understanding from an agent system into a single generative Video-LLM. Compared with conventional instruction tuning, the distilled model answers compositional questions more reliably and verbalizes grounded rationales.

1. Introduction

Recent Video-LLMs have made rapid progress on general video understanding, propelled by instruction-tuned multimodal models such as Video-ChatGPT [22], Video-LLaVA [20], LLaMA-VID [19], LLaVA-NeXT-Video [45], and VideoLLaMA2 [4]. However, benchmarks centered on situated, causal, and compositional reasoning still expose a persistent gap between recognition and reasoning. Datasets such as STAR [35], NExT-QA [36], AGQA [14], ANetQA [43], CLEVRER [40], MVBench [17], VideoMME [9], Perception-Test [29], and VSI-Bench [37] demand not only answer prediction but also localization of the relevant moments, actors, and objects.

Two broad solution families have emerged. End-to-end Video-LLMs are efficient and easy to deploy, but their reasoning is opaque and their errors are difficult to diagnose. In contrast, modular and agentic systems decompose a query into smaller operations and solve them with specialized tools [5, 7, 24, 38]. These systems are more interpretable and often more controllable, but they are slow, memory intensive, and brittle to tool or planning failures. The central question of this paper is whether we can keep the structured reasoning benefits of agents while recovering the simplicity and efficiency of a single Video-LLM.

We answer this question with *Agent-of-Thoughts Distillation* (AoTD). The key idea is to use an agent to generate tool-grounded reasoning traces and distill them into a generative Video-LLM through supervised instruction tuning. Our agent receives a video and question, generates executable Python-style programs over a predefined API, and solves sub-problems with specialized models for question decomposition, temporal grounding, object detection, action recognition, and answer selection. The resulting execution traces contain temporally localized segments, object boxes, intermediate summaries, and final answers. We rewrite these traces as natural-language chains of thought, filter them for step-by-step usefulness, and train a Video-LLM on both direct-answer and rationale-generation prompts. The output is a single model that does not invoke external tools during inference, yet inherits part of the agent's reasoning structure.

This design is motivated by two observations. First, the agent's intermediate outputs provide grounded supervision that ordinary answer-only instruction data lacks. Second, reasoning traces are only useful if they are both correct and transferable, so they should be filtered rather than treated as free supervision. The resulting distilled model improves over matched instruction tuning on both multiple-choice and open-ended benchmarks, while remaining substantially cheaper than running the full agent pipeline.

Our contributions are threefold. First, we propose AoTD, a practical distillation framework that transfers agent-generated spatial-temporal reasoning into a single Video-LLM. Second, we build an agent pipeline that selects the best available tool per sub-task, converts execution traces into natural-language rationales, and filters them for supervision quality. Third, we show that the resulting ratio-

nale supervision improves benchmark accuracy, transfers to multiple Video-LLM backbones, and yields grounded rationales at a fraction of the cost of the original agent system.

2. Related Work

Video-LLMs and video question answering. Instruction-tuned multimodal models have quickly become the standard recipe for video understanding [1, 3, 4, 19, 20, 22, 44, 45]. At the same time, recent benchmarks have shifted attention from generic captioning to temporal, causal, and compositional reasoning [9, 14, 17, 29, 35–37, 43]. Our work targets this harder regime: we keep the end-to-end deployment advantages of Video-LLMs, but change the supervision signal from answer-only tuning to tool-grounded rationale distillation.

Modular and agentic video reasoning. Several works improve interpretability by decomposing video reasoning into explicit intermediate steps. SeViLA chains localization and answering within a single image-language model [41]. MoReVQA [24], ProViQ [5], VideoAgent [7], DoraemonGPT [38], and Video-of-Thought [8] use planners, external tools, or structured memories to answer complex video queries. These methods demonstrate that modularity is useful, but they retain multi-model inference at test time. AoTDdiffers in goal: we use the agent only during data construction, then distill its reasoning into a single generative Video-LLM.

Chain-of-thought and rationale distillation. Chain-of-thought supervision has proved effective for language models [33, 39] and has been extended to multimodal reasoning through prompting and rationale datasets [11, 26, 31, 46]. Closer to our setting, VPD [15] and FACT [10] distill programmatic or executable reasoning into image-based multimodal models. Video-STaR [47] leverages existing labels for video instruction tuning through self-training. Our contribution is complementary: we focus on video reasoning, construct rationales from agent execution traces that contain explicit temporal and spatial evidence, and study both rationale filtering and distillation transfer across Video-LLM backbones.

3. Agent-of-Thoughts Distillation

Problem setup. Given a video v and question q, standard VideoQA predicts an answer y. Our goal is to train a Video-LLM that predicts y while also internalizing an interpretable reasoning process. Instead of collecting human-written rationales, we construct supervision from an agent system that can solve the question by invoking specialized tools. The distilled model is trained once and then used without external tools at test time.

Sub-task	Model name	Metric	Number (%)
question decomposition	CodeQwen1.5-Chat (7B) [2]	Acc	52.7
	GPT-3.5-Turbo [27]		73.1
	DeepSeek-Coder-Instruct (6.7B) [6]		**85.7**
object detection	OWL-ViT v1 [23]	IoU	47.3
	GLIP [18]		58.9
	OWL-ViT v2 [25]		**63.0**
temporal grounding	LITA (13B) [16]	IoU / Recall	11.7 / 20.2
	TimeChat (7B) [30]		13.9 / 23.1
	UniVTG [21]		**24.7 / 35.3**
action recognition	InternVideo2 (1B) [32]	Top1-Acc	7.6
	Open-VCLIP [34]		8.9
	LLaVA-NeXT-Video-DPO (7B) [45]		**18.2**
question answering	LLaMA-VID (7B) [19]	Acc	43.5
	SeViLA [41]		46.5
	LLaVA-NeXT-Video-DPO (7B) [45]		**53.4**

Table 1. Sub-tasks definition and evaluation results. We choose 3 model candidates for each sub-task and evaluate them in STAR training set with the corresponding metrics. Models with best performance are placed at the bottom of each column.

Sub-task library and model selection. The agent operates over a small API exposed in the planning prompt, including routines for querying objects, querying actions, filtering frames by object or action, trimming clips, extracting person-centered crops, selecting an answer from candidate options, testing object existence, and summarizing a clip. To instantiate this API, we consider five core sub-tasks: question decomposition, object detection, temporal grounding, action recognition, and question answering. We evaluate three candidate models per sub-task on the STAR training split and choose the strongest tool according to the task-specific metric. Table 1 shows the selected models: DeepSeek-Coder-Instruct for question decomposition, OWL-ViT v2 for object detection, UniVTG for temporal grounding, and LLaVA-NeXT-Video-DPO for both action recognition and one-hop question answering. This agent composition lets us mix best-in-class capabilities rather than relying on a single monolithic model.

Program generation and execution. For each training question, the planner generates a Python-style `execute_command` function over the API. DeepSeek-Coder produces the program, which is then executed over uniformly sampled video frames and tool outputs. The system records the program itself, the execution trace, and the final answer. These traces are grounded: temporal grounding returns intervals, object detection yields boxes, and the answering tool consumes the retrieved evidence. Figure 2 illustrates the full pipeline, and Figure 4 shows examples of the final distilled rationales.

Trace-to-rationale conversion and filtering. Raw traces are not directly suitable as training targets. We therefore post-process them in two stages. First, we keep only exe-

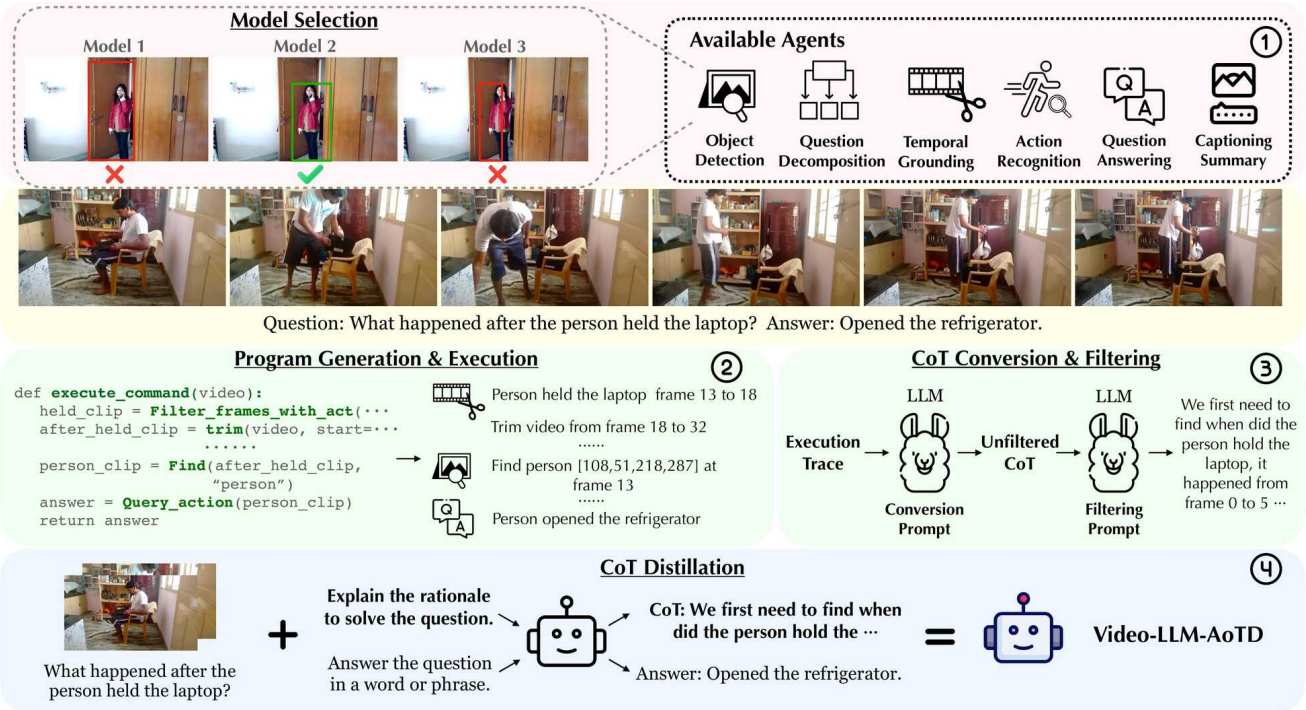

Figure 2. Overview of AoTD. We first select the strongest model for each sub-task and use a code-generating planner to produce executable programs over these tools. The resulting execution traces are converted into natural-language chains of thought and filtered for quality. Finally, we fine-tune a single Video-LLM with two prompt formats: direct answering and rationale generation.

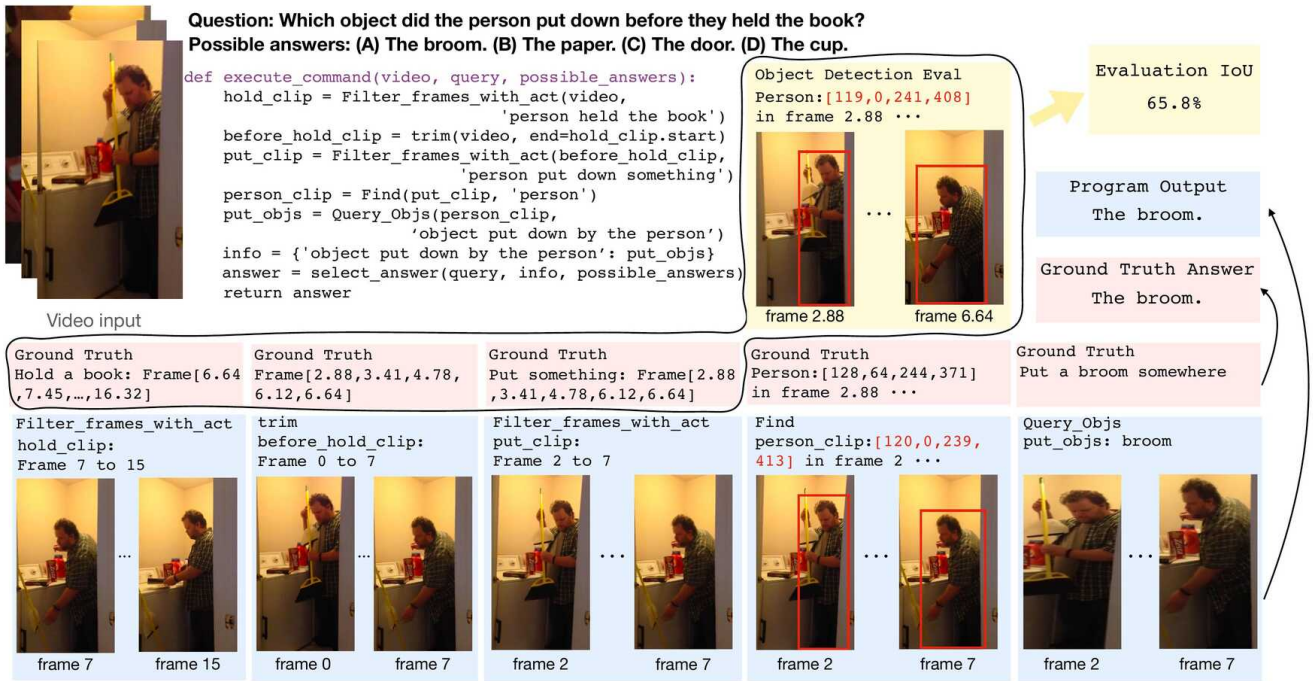

Figure 3. Example execution flow from the agent system. The planner emits an executable program, tools return temporal spans and visual evidence, and the verified trace is rewritten into the rationale used for distillation.

cutions whose program output matches the ground-truth answer and discard ambiguous labels such as "both right" and "both false," following the released preprocessing scripts. Second, we prompt an instruction-tuned language model

Dataset	Description	# Labels	# CoTs
AGQA	Compositional	25.0K	5.4K
ANetQA	Compositional	25.0K	3.6K
STAR	Compositional	45.7K	11.2K
NExT-QA	Temporal & Causal	34.1K	12.1K
CLEVRER	Spatial & Temporal	21.0K	-
EgoQA	Ego-centric	7.8K	-
Total		**158.6K**	**32.3K**

Table 2. Dataset statistics. The column "# Labels" indicates the number of VideoQA pairs, including the video, query, candidate answers for multiple-choice settings, and the correct answer. "# CoTs" denotes the number of distilled rationale candidates generated by the agent system.

to rewrite the trace into a natural-language rationale. The transfer prompt explicitly asks the model to preserve the temporal and bounding-box evidence instead of summarizing it away. A second filter prompt retains only rationales that are both step-by-step and helpful for solving the question. The resulting supervision is intentionally conservative: it favors fewer but cleaner rationales over maximizing raw data volume.

Distillation data. Table 2 summarizes the final corpus. Across STAR, NExT-QA, AGQA, and ANetQA, the agent constructs and filters 32.3K rationale examples from 158.6K labeled VideoQA pairs. We use the full labeled datasets for answer supervision and the retained rationales for rationale supervision. Table 3 lists the train/evaluation split. Training data covers both multiple-choice and open-ended VideoQA, while evaluation additionally includes MVBench, VideoMME, Perception-Test, VSI-Bench, ActivityNet-QA, and Video-ChatGPT.

Dataset	Size		Type	Train	Eval
	train	eval			
MC-VQA					
STAR [35]	45.7K	7.1K	Compositional	✓	✓
NExT-QA [36]	34.1K	5.0K	Temporal & Causal	✓	✓
CLEVRER [40]	21.0K	-	Spatial-temporal	✓	✗
Perception-Test [29]	-	11.5K	General	✗	✓
MVBench [17]	-	4.0K	General	✗	✓
VideoMME [9]	-	2.7K	General	✗	✓
VSIBench [37]	-	5.0K	spatial-temporal	✗	✓
OE-VQA					
AGQA [14]	25.0K	2.0K	Compositional	✓	✓
ANetQA [43]	25.0K	2.0K	Compositional	✓	✓
EgoQA [13]	7.8K	-	Ego-centric	✓	✗
Activitynet-QA [42]	-	8.0K	General	✗	✓
Video-ChatGPT [22]	-	3.0K	General	✗	✓

Table 3. Training and evaluation dataset statistics.

Dual-prompt distillation. We fine-tune a LLaVA-NeXT-Video 7B model with two complementary prompt formats. The answer prompt asks for only the option letter for multiple-choice questions or a short phrase for open-ended questions. The rationale prompt changes only the suffix instruction and asks the model to "Explain the rationale to answer the question." Training alternates between answer targets and rationale targets under the same autoregressive objective conditioned on the corresponding suffix prompt. In the released training code, videos are uniformly sampled to 32 frames, the maximum sequence length is 8192 tokens, the batch size is 4, the learning rate is 4×10^{-5}, and training runs for one epoch with a frozen visual tower.

4. Experiments

Benchmarks and metrics. For multiple-choice VideoQA, we report accuracy on MVBench, VideoMME, STAR, NExT-QA, Perception-Test, and VSI-Bench. For open-ended VideoQA, we report accuracy and score on ANetQA, AGQA, and ActivityNet-QA, plus the five Video-ChatGPT dimensions: correctness, detail, context, temporal, and consistency. We follow the official benchmark protocols and compare against both proprietary systems and open-source Video-LLMs. Our main paired baseline is LLaVA-NeXT-Video-Instruct, which shares the same backbone but is trained without distilled chains of thought.

Main results on multiple-choice VideoQA. Table 4 shows that AoTDimproves the matched instruction-tuned baseline on all six multiple-choice benchmarks. Relative to LLaVA-NeXT-Video-Instruct, AoTDgains +2.2 points on MVBench, +1.8 on VideoMME, +2.1 on STAR, +0.5 on NExT-QA, +0.3 on Perception-Test, and +2.1 on VSI-Bench. These improvements are strongest on benchmarks with explicit compositional or spatial-temporal reasoning demands, namely MVBench, STAR, and VSI-Bench. AoTDalso outperforms the reproduced zero-shot LLaVA-NeXT-Videobaseline by large margins, indicating that the gain comes from rationale-aware distillation rather than from the backbone alone.

Main results on open-ended VideoQA. Table 5 shows consistent gains over LLaVA-NeXT-Video-Instruct on all three open-ended accuracy benchmarks: ANetQA rises from 47.1/3.1 to 53.9/3.4, AGQA from 59.3/3.4 to 60.9/3.6, and ActivityNet-QA from 50.0/3.3 to 53.2/3.4. On the Video-ChatGPT evaluation, AoTDimproves correctness, detail, context, and consistency over the instruction-tuned baseline, but not every dimension improves uniformly: the temporal sub-score changes from 2.42 to 2.41. This small drop is important because it shows that rationale distillation does not automatically solve all aspects of free-form temporal description, even when overall answer quality improves.

Model	MVBench (Acc.)	VideoMME (Acc.)	STAR (Acc.)	NExT-QA (Acc.)	Perception-Test (Acc.)	VSIBench (Acc.)
Proprietary Models						
Gemini 1.5 Pro [12]	-	**75.0**	-	-	-	**45.4**
GPT4-V [28]	43.7	59.9	-	-	-	-
Open-source Models						
LLaMA-VID (7B) [19]	41.9	25.9	-	-	44.6	-
Video-LLaVA (7B) [20]	41.0	39.9	-	-	44.3	-
VideoChat2 (7B) [17]	51.1	33.7	59.0*	68.6*	47.3	-
VideoLLaMA2 (7B) [4]	53.4	45.1	58.5*	62.3*	49.6	-
LLaVA-NeXT-Video (7B) [45]	46.5*	41.0*	52.4*	61.6*	47.5*	19.7*
LLaVA-NeXT-Video-Instruct (7B)	53.4	43.2	72.2	77.1	50.3	26.7
LLaVA-NeXT-Video-AoTD (7B)	**55.6**	45.0	**74.3**	**77.6**	**50.6**	28.8

Table 4. Comparison with Video-LLMs on MC-VQA benchmarks. LLaVA-NeXT-Video-AoTD consistently improves over the matched instruction-tuned baseline and attains the best open-source result on all six benchmarks. * denotes results reproduced by ourselves. Results without this mark are taken from [4, 37].

Model	ANetQA (Acc./Score)	AGQA (Acc./Score)	Video-ChatGPT (Score)					ActivityNet (Acc./Score)
			Corr.	Deta.	Cont.	Temp.	Cons.	
Proprietary Models								
Gemini 1.5 Pro [12]	-	-	-	-	-	-	-	56.7/-
GPT4-V [28]	-	-	**4.09**	**3.88**	**4.37**	**3.94**	**4.02**	**59.5/-**
Open-Source Models								
VideoLLaMA (7B) [44]	-	-	1.96	2.18	2.16	1.82	1.79	12.4/1.1
Video-ChatGPT (7B) [22]	-	-	2.50	2.57	2.69	2.16	2.20	35.2/2.7
LLaMA-VID (7B) [19]	-	-	2.96	3.00	3.53	2.46	2.51	47.4/3.3
Video-LLaVA (7B) [20]	-	-	2.87	2.94	3.44	2.45	2.51	45.3/3.3
VideoChat2 (7B) [17]	-	-	3.02	2.88	3.51	2.66	2.81	49.1/3.3
VideoLLaMA2 (7B) [4]	-	-	3.09	3.09	3.68	2.63	3.25	49.9/3.3
LLaVA-NeXT-Video (7B) [45]	46.4/3.3*	27.4/2.2*	3.26*	3.22*	3.77*	2.47*	2.99*	54.3/3.2*
LLaVA-NeXT-Video-Instruct (7B)	47.1/3.1	59.3/3.4	2.96	2.81	3.35	2.42	2.82	50.0/3.3
LLaVA-NeXT-Video-AoTD (7B)	**53.9/3.4**	**60.9/3.6**	3.11	3.00	3.60	2.41	2.91	53.2/3.4

Table 5. Comparison with Video-LLMs on OE-VQA benchmarks. LLaVA-NeXT-Video-AoTD improves over the matched instruction-tuned baseline on all open-ended benchmarks. * denotes results reproduced by ourselves. Results without this mark are taken from [4].

Inference efficiency. The main practical advantage of distillation is that the test-time agent disappears. Table 6 compares the deployed AoTDmodel with the original agent system. Distillation reduces latency from 47.93 s to 10.58 s, cuts memory from 65.34 GB to 18.21 GB, and lowers compute from 30.98 to 13.53 TFLOPs. In other words, AoT-Dretains much of the agent's reasoning benefit while turning a multi-model inference pipeline into a single forward pass.

Model	Time (s) ↓	Memory (GB) ↓	TFLOPs ↓
Agent-based system	47.93	65.34	30.98
LNV-AoTD	**10.58**	**18.21**	**13.53**

Table 6. Latency and computation comparison.

5. Analysis

How much does rationale filtering matter? Not every generated rationale is useful. Table 7 compares AoTDwith and without the final CoT filtering stage on the LLaVA-NeXT-Videobackbone. Filtering improves MVBench from 53.7 to 55.6, STAR from 73.3 to 74.3, and AGQA from 59.5/3.5 to 60.9/3.6. This supports preferring verified and transferable rationales over unfiltered agent traces. The same table also tests transferability to other backbones. AoTD-style rationale supervision improves LLaVA-OneVision, VideoLLaMA2, and Qwen2-VL as well, suggesting that the method is not tied to one architecture.

Model	Filtering	MVBench (Acc.)	STAR (Acc.)	AGQA (Acc. / Score)
LNV-AoTD	✗	53.7	73.3	59.5/3.5
LNV-AoTD	✓	**55.6**	**74.3**	**60.9/3.6**
Onevision-Instruct	-	59.2	75.8	65.6/3.7
Onevision-AoTD	✓	**60.5**	**76.6**	**65.7/3.7**
VideoLLaMA2-Instruct	-	54.9	69.2	56.0/3.5
VideoLLaMA2-AoTD	✓	**56.0**	**71.1**	**57.2/3.5**
Qwen2-VL-Instruct	-	65.6	71.4	59.8/3.6
Qwen2-VL-AoTD	✓	**66.5**	**73.1**	**61.2/3.7**

Table 7. Ablation results of CoT filtering and transferability.

Model	Temporal Grounding		Spatial Grounding
	IoU (%)	Recall (%)	IoU (%)
UniVTG	**22.8**	31.0	-
OWL-ViT v2	-	-	**64.7**
LNV-Instruct	✗	✗	✗
LNV-AoTD	21.7	**34.0**	45.2

Table 8. Temporal and spatial abilities evaluation results.

Does the distilled model retain grounding ability? Answer accuracy alone does not show whether the model internalizes the agent's spatial-temporal evidence. We therefore evaluate the generated rationales themselves. Table 8 shows that AoTDreaches 21.7 temporal IoU and 34.0 temporal recall, compared with 22.8 and 31.0 for the specialist UniVTG temporal grounding model. For spatial grounding, AoTDattains 45.2 IoU, well below the dedicated OWL-ViT v2 detector at 64.7 IoU but substantially more informative than the answer-only instruction-tuned baseline, which does not yield grounded rationales. This pattern suggests that temporal reasoning transfers more faithfully than precise object localization, which remains the harder capability to absorb into a single generative model.

Qualitative behavior. Figure 4 illustrates the type of reasoning learned by AoTD. The distilled model refers to time ranges and localized regions that support the answer. These rationales are not perfect substitutes for specialist tools, as reflected by the lower spatial IoU in Table 8, but they show that the model has learned a more structured internal decision process than conventional answer-only instruction tuning.

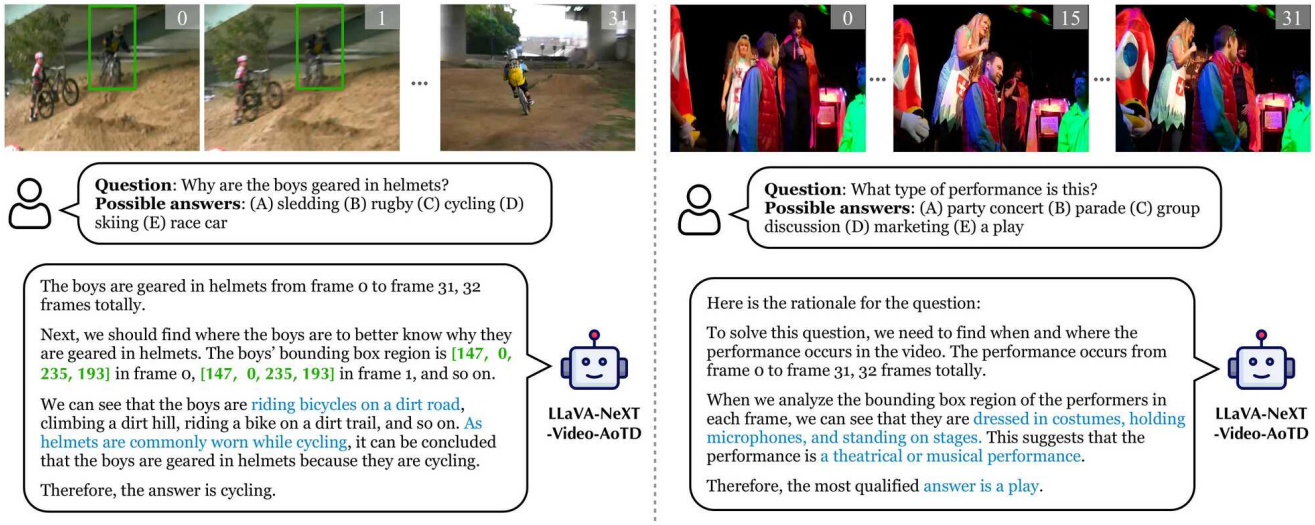

Figure 4. Visualization of rationales generated by the distilled model. The outputs explicitly mention temporal spans and bounding boxes, showing that AoTDlearns to verbalize grounded evidence rather than only predicting the final answer.

6. Conclusion

We presented AoTD, a distillation framework that transfers tool-grounded, multi-step reasoning from an agent system into a single Video-LLM. The method builds an agent out of strong specialized models, rewrites successful execution traces into natural-language rationales, filters them for supervision quality, and fine-tunes a Video-LLM with both answer and rationale targets. Experiments show that this recipe consistently improves a strong LLaVA-NeXT-Video 7B baseline on both multiple-choice and open-ended VideoQA while greatly reducing inference cost relative to the original agent. The analysis also clarifies the current boundary of the approach: temporal reasoning transfers well, whereas precise spatial grounding remains only partially distilled.

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